

- 1** A graphic designer creates a square logo of side length L .
The logo is first reduced in size so that its side length decreases by 25%.
Realising the logo is too small, the designer attempts to restore it to its original size by increasing the side length of the reduced logo by 25%.
The area of this final square logo is 31 cm^2 smaller than the area of the original square logo.
What is the perimeter of the original square logo?
- A** 32 cm
B 48 cm
C 60 cm
D 64 cm
E 256 cm
- 2** A set of numbers is $S = \{1, 2, 3, 4, 5\}$. A number is chosen at random from the set S three times, with replacement.
What is the probability that the product of the three chosen numbers is an even number?
- A** $\frac{8}{125}$
B $\frac{27}{125}$
C $\frac{54}{125}$
D $\frac{98}{125}$
E $\frac{117}{125}$
F $\frac{1}{2}$
- 3** The value of an asset increases by 60%. The new value then decreases by 25%. The final value of the asset is 1080.
What was the initial value of the asset?
- A** 540
B 800
C 864
D 900
E 1296

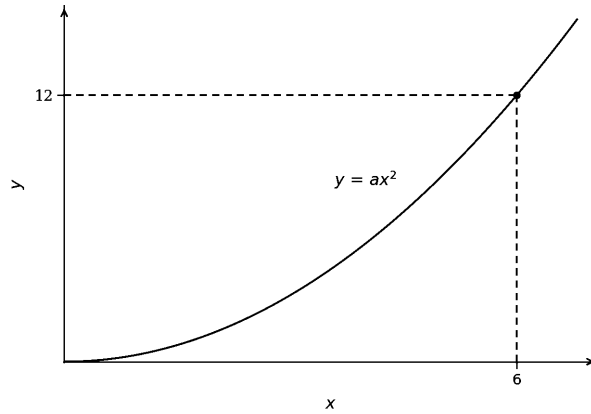
- 4 A car travels a certain distance at 30 km/h. It then travels twice that distance at 90 km/h. What is the average speed of the car for the whole journey?
- A 40 km/h
 - B 45 km/h
 - C 54 km/h
 - D 60 km/h
 - E 70 km/h
- 5 A quantity x is increased by 150% to give quantity y . Quantity z is $q\%$ less than y . Quantity z is also 50% less than x . What is the value of q ?
- A 20
 - B 40
 - C 80
 - D 100
 - E 200

- 6 The diagram shows the graph of $y = ax^2$ for $x \geq 0$, where a is a constant. The curve passes through the point $(6, 12)$.

The variable x is inversely proportional to t .

When $t = 4$, $x = 3$.

Which of the following gives an expression for y in terms of t ?



- A** $y = \frac{4}{t}$
- B** $y = \frac{4}{t^2}$
- C** $y = \frac{3}{16t^2}$
- D** $y = \frac{48}{t^2}$
- E** $y = \frac{288}{t^2}$
- F** $y = \frac{432}{t^2}$
- G** $y = \frac{3t^2}{16}$
- H** $y = 48t^2$
- 7 A composite cylindrical rod has length L and total radius $2r$. It consists of a solid inner core of radius r and a surrounding outer sleeve. The core is made of a material with density 3ρ and the sleeve is made of a material with density ρ .
- Which expression gives the total mass of the rod?
- A** $4\pi r^2 L \rho$
- B** $6\pi r^2 L \rho$
- C** $7\pi r^2 L \rho$
- D** $8\pi r^2 L \rho$
- E** $10\pi r^2 L \rho$
- F** $12\pi r^2 L \rho$

- 8 A digital file of size 120 Megabytes is to be transmitted over a network. The transmission protocol requires that for every 15 parts of data, 1 part of overhead is added.

The network has a constant transmission speed of 32 Megabits per second.

Assuming 1 Megabyte = 10^6 bytes, and 1 byte = 8 bits, what is the total time required to transmit the file and its overhead?

- A 0.5 seconds
- B 2 seconds
- C 4 seconds
- D 30 seconds
- E 32 seconds
- F 480 seconds

- 9 Pipe P can fill a tank in 10 hours. Pipe Q can fill the same tank in T hours.

Pipe P works alone for 2 hours to begin filling the empty tank. Then pipe Q is also opened, and working together, they take a further 4 hours to fill the tank.

What is the value of T ?

- A 5
- B 7.5
- C 8
- D 10
- E 15
- F 20

- 10 Two quadratic equations of the form $x^2 + 2bx + c = 0$ are formed.

For each equation, the coefficients b and c are chosen independently and uniformly at random from the set $\{1, 2, 3\}$.

What is the probability that at least one of the two equations has real roots?

- A $\frac{4}{81}$
- B $\frac{28}{81}$
- C $\frac{49}{81}$
- D $\frac{7}{9}$
- E $\frac{8}{9}$
- F $\frac{77}{81}$

- 11** The finite region R is bounded by the four lines with equations:

$$3x + 4y = 22$$

$$3x - 4y = 14$$

$$3x + 4y = -2$$

$$3x - 4y = -10$$

What is the area of R ?

- A** 12
 - B** 20
 - C** 24
 - D** 25
 - E** 48
- 12** The positive integers x, y, z have no common factor strictly greater than 1. They satisfy the relationship:

$$3x : 4y : 5z = 2 : 3 : 4$$

Given that k is a constant such that $kx + y - z = 77$, what is the value of k ?

- A** 2
- B** 26
- C** 27
- D** 38
- E** 39
- F** 40

- 13** Let P be a quadrilateral that has rotational symmetry of order greater than 1, but no line symmetry.
Let Q be a quadrilateral that has one or more lines of symmetry, but has rotational symmetry of order 1.

Which of the following is a possible pair (P , Q)?

(Assume any named shape is a non-special case, e.g. a 'rectangle' is not a square.)

- A** (Parallelogram, Kite)
- B** (Kite, Parallelogram)
- C** (Rectangle, Kite)
- D** (Parallelogram, Rhombus)
- E** (Rhombus, Rectangle)
- F** (Square, Isosceles Trapezium)

- 14** A curve is given by the equation $y = (x - k)(x^2 - 4)$, where k is a real constant such that $k > 1$.
A dataset consists of the y -intercept and all the x -intercepts of the curve.

The mean of this dataset is equal to its median.

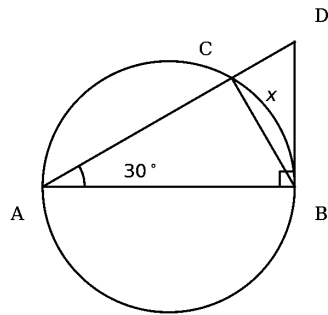
What is the value of k ?

- A** $\frac{2}{9}$
- B** $\frac{4}{7}$
- C** $\frac{4}{3}$
- D** $\frac{8}{5}$
- E** $\frac{8}{3}$
- F** 6

- 15 The diagram shows a circle with diameter AB .

The straight line AD passes through point C on the circumference of the circle, and the line BD is tangent to the circle at B .

Given that $\angle CAB = 30^\circ$ and the length of BC is x , which of the following is a correct expression for the length of CD in terms of x ?



[diagram not to scale]

- A $\frac{1}{3}x$
- B $\frac{1}{2}x$
- C $\frac{\sqrt{3}}{6}x$
- D $\frac{\sqrt{3}}{3}x$
- E $\frac{\sqrt{3}}{2}x$
- F $\frac{2\sqrt{3}}{3}x$
- G $\sqrt{3}x$
- H $2\sqrt{3}x$

- 16** Two cones have the same volume. The height of the second cone is twice the height of the first cone.

You may use the formula

$$V = \frac{1}{3}\pi r^2 h.$$

If the base radius of the first cone is r , what is the base radius of the second cone?

- A** $\frac{r}{\sqrt{2}}$
- B** $\frac{r}{2}$
- C** $\frac{r}{4}$
- D** $r\sqrt{2}$
- E** $2r$
- F** $4r$

- 17** The curve C has equation $y = 6x - x^2$.

The point M is the maximum point of C .

The points P and Q have position vectors $\vec{OP}$ and $\vec{OQ}$ such that:

$$3\vec{OM} = \vec{OP} + 2\vec{OQ}$$

where O is the origin.

Given that P has coordinates $(7, 3)$, what is the distance from Q to the point where C intersects the positive x -axis?

- A** 13
- B** $\sqrt{13}$
- C** $\sqrt{73}$
- D** $\sqrt{145}$
- E** $\sqrt{229}$
- F** $4\sqrt{37}$

- 18** Liquid X has a density of 4.
Liquid A has a density of 5.
Liquid B has a density of 2.

A mixture, Y, is created by combining liquids A and B. A new blend is then formed by mixing liquid X and mixture Y.

In this final blend, the ratio of the volume of X to the volume of Y is 1 : 3.

The ratio of the mass of X to the mass of Y is 2 : 5.

What is the ratio of the volume of liquid A to the volume of liquid B in mixture Y?

- A** 1 : 1
- B** 2 : 1
- C** 4 : 5
- D** 5 : 4
- E** 14 : 1
- F** 4 : 25

- 19** A fair six-sided die is rolled.
If the number rolled is a multiple of 3, a fair coin is tossed 3 times.
Otherwise, the coin is tossed 2 times.
What is the probability of obtaining exactly 2 heads?

- A** $\frac{5}{24}$
- B** $\frac{13}{48}$
- C** $\frac{7}{24}$
- D** $\frac{5}{16}$
- E** $\frac{1}{3}$
- F** $\frac{3}{8}$

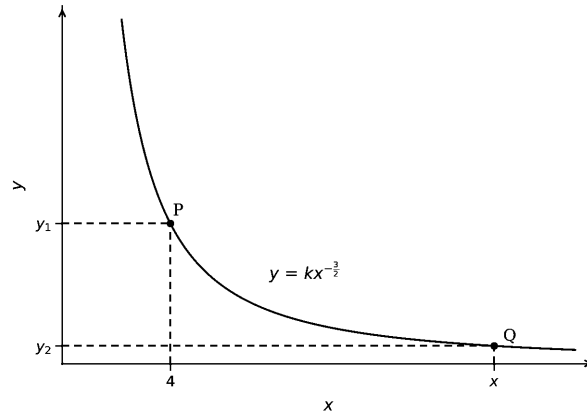
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The diagram shows part of the curve with equation $y = kx^{-\frac{3}{2}}$, where k is a positive constant and $x > 0$.

The points $P(4, y_1)$ and $Q(x, y_2)$ lie on the curve.

The value of y_2 is 87.5% less than the value of y_1 .

What is the value of x ?



- A 8
- B $8\sqrt{2}$
- C 16
- D 32
- E 64

- 21 Three fair, six-sided dice are rolled. Let p be the probability that the highest value shown is 4. Let q be the probability that the lowest value shown is 2. What is the value of $q - p$?

- A 0
- B $\frac{7}{72}$
- C $\frac{1}{9}$
- D $\frac{37}{216}$
- E $\frac{61}{216}$

- 22** A rectangle has vertices at the origin O, and at the points A(x , 0), B(x , y) and C(0, y), where x and y are positive integers.

The area of the rectangle OABC is 360.

The gradient, m , of the diagonal OB satisfies the equation:

$$4x - \frac{45}{m} = 30$$

Given that $m < 1$, what is the perimeter of the rectangle?

- A** 38
B 42
C 76
D 78
E 84
F 92
- 23** A composite solid sphere of outer radius $2r$ consists of a central spherical core of radius r and density ρ_c , surrounded by a shell of density ρ_s .
The mean density of the composite sphere is $2\rho_s$.
What is the ratio $\frac{\rho_c}{\rho_s}$?
- A** 5
B 7
C 8
D 9
E 15
- 24** A data set S contains n distinct integers, where $n \geq 2$. Let μ be its mean, m its median and r its range. A further integer data value k is appended to the data set; repeated values are allowed after this addition.
Which statements must be true?
1. If $k > \mu$, the new mean is greater than μ .
 2. If $k > r$, the new range is greater than r .
 3. If $k < m$, the new median is less than m .
- A** 1 only
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only
E 1, 2 and 3

- 25** For a regular polygon with n sides, the angle between a side extended and the adjacent side is $(k - 20)^\circ$, where k is a constant.

Which of the following expressions gives the value of n ?

- A** $\frac{180}{k - 20}$
B $\frac{360}{k - 20}$
C $\frac{180}{200 - k}$
D $\frac{360}{200 - k}$

- 26** The variables x and y are such that the graph of y against x is a straight line through the origin. When $x = 2$, $y = 8$.

The constants p and q are given by $p = 5 \times 10^7$ and $q = 2 \times 10^{-3}$.

A quantity M is defined as:

$$M = \frac{\left(\frac{y}{q}\right)}{\left(\frac{x}{p}\right)}$$

What is the value of M ?

- A** 1×10^5
B 6.25×10^9
C 2.5×10^{10}
D 1×10^{11}
E 1.6×10^{-10}
F 6.25×10^3

- 27** The real number x satisfies the equation:

$$\frac{x - \sqrt{3}}{x + \sqrt{3}} = 2 - \sqrt{3}$$

What is the value of x ?

- A** -3
B -1
C 1
D 2
E 3